

(centipedes and millipedes); and **pancrustaceans** (a recently defined, diverse group that includes insects as well as lobsters, shrimp, barnacles, and other crustaceans).

Chelicerates

Chelicerates (clade Chelicerata) are named for clawlike feeding appendages called **chelicerae**, which serve as pincers or fangs. Chelicerates lack antennae, and most have simple eyes (eyes with a single lens).

The earliest chelicerates were **eurypterids**, or water scorpions. These marine and freshwater predators grew up to 3 m long; it is thought that some species could have walked on land, much as land crabs do today. Most of the marine chelicerates, including all of the eurypterids, are extinct. Among the marine chelicerates that survive today are the sea spiders (pycnogonids) and horseshoe crabs (**Figure 33.31**).

The bulk of modern chelicerates are **arachnids**, a group that includes scorpions, spiders, ticks, and mites (**Figure 33.32**). Nearly all ticks are bloodsucking parasites that live on the body surfaces of reptiles or mammals. Parasitic mites live on or in a wide variety of vertebrates, invertebrates, and plants.

Arachnids have six pairs of appendages: the chelicerae; a pair of appendages called *pedipalps* that function in sensing, feeding, defense, or reproduction; and four pairs of walking legs. Spiders use their fang-like chelicerae, which are equipped with poison glands, to attack prey. As the chelicerae pierce the prey, the spider secretes digestive juices onto the prey's torn tissues. The food softens, and the spider sucks up the liquid meal. In most spiders, gas exchange is carried out by **book lungs**, stacked platelike structures contained in an internal chamber (**Figure 33.33**). The extensive surface area of these respiratory organs enhances the exchange of O₂ and CO₂ between the hemolymph and air.

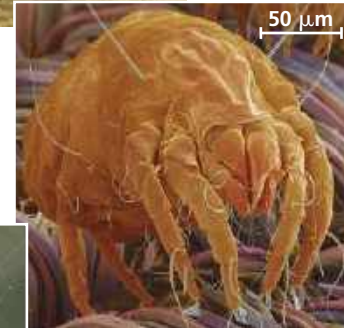
A unique adaptation of many spiders is the ability to catch insects by constructing webs of silk, a liquid protein produced

▼ Figure 33.31 Horseshoe crabs (*Limulus polyphemus*).

Common on the Atlantic and Gulf coasts of the United States, these "living fossils" have changed little in hundreds of millions of years. They are surviving members of a rich diversity of chelicerates that once filled the seas.



▲ Scorpions have pedipalps that are pincers specialized for defense and the capture of food. The tip of the tail bears a poisonous stinger.



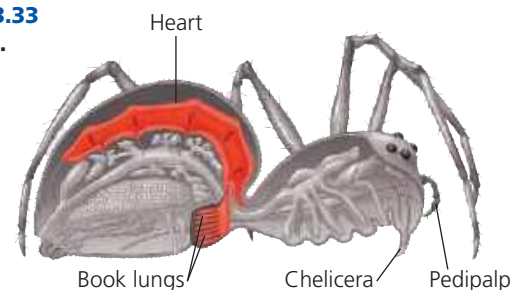
▲ Dust mites are ubiquitous scavengers in human dwellings but are harmless except to those people who are allergic to them (colorized SEM).



◀ Web-building spiders are generally most active during the daytime.

▲ **Figure 33.32 Arachnids.**

► Figure 33.33 Book lungs.



by specialized abdominal glands. The silk is spun by organs called spinnerets into fibers that then solidify. Each spider engineers a web characteristic of its species and builds it perfectly on the first try, indicating that this complex behavior is inherited. Various spiders also use silk in other ways: as droplines for rapid escape, as a cover for eggs, and even as "gift wrap" for food that males offer females during courtship. Many small spiders also extrude silk into the air and let themselves be transported by wind, a behavior known as "ballooning."

Myriapods

Millipedes and centipedes belong to the clade Myriapoda (**Figure 33.34**). All living myriapods are terrestrial. The myriapod head has a pair of antennae and three pairs of